



U.S. General Services
Administration



Agentic AI Book Club - Session 1

Reviewing Chapter 1.1 A & B

US General Service Administration
AI Community of Practice - March 22026

Agenda

- Housekeeping
 - Chapter review
 - Break - Reflection
 - Quiz review
 - Intro to next week's chapters
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House Keeping

- **Agentic AI is an early and rapidly evolving knowledge domain**
 - If you missed Chapter 1.1A & B, move on to 1.2.
 - Activities including quizzes should be done in your own time. Quiz form links can be accessed from personal devices
 - “Tracking code” is your email. If you see quiz questions with square boxes, they are multi-select questions.
 - Python coding is helpful but not mandatory. Code examples are in https://github.com/GSA/MasteringAgenticAISystems_supplementals/tree/master/code_examples
 - Week 1 slides will be included in week 2 materials and so on
 - Please do quizzes and build confidence towards taking real certification exams! (any one of the 5)
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What Makes Agent UIs Different?

Traditional UIs: users drive every decision — forms, buttons, workflows.

Agent UIs: agents make decisions — users oversee, review, and intervene.

The user's role shifts from operator to overseer.

This shift requires new design patterns beyond classic usability (clarity, consistency, feedback, etc.)



The Trust Equation

Research identifies two strongest predictors of user trust in AI systems:

- Transparency — users understand what the agent is doing and why
- Control — users can intervene, modify, or override agent actions

The relationship is multiplicative, not additive:

- Transparency without control → informed frustration
- Control without transparency → anxious over-monitoring

Both are required, calibrated to decision impact.



Four Failure Modes of Poor Agent UI

1. Trust loss — opaque behavior, unexplained conclusions
2. Safety risk — high-stakes actions execute without human review
3. Cognitive overload — every tool call and API response displayed by default
4. System abandonment — users return to manual processes silently

The most dangerous failure is silent abandonment: deployment numbers look fine, usage has already collapsed.



Agentic UI Design Principles?

Agentic UI Design Principles

01.

Progressive Disclosure

respect cognitive limits,
maintain transparency

02.

Transparency and Explainability

make agent behavior
understandable

03.

User Control and Intervention

maintain meaningful human
authority

04.

Error Communication and Recovery

turn failures into learning
opportunities

05.

Context Awareness and Continuity

maintain conversation
coherence

*Note

Exceptions

Nothing is rigid with AI. You
must be willing to make
well-designed exceptions

Control Spectrums and Patterns

1

Low

Notification. Agent acts;
user sees post-facto
summary with undo

2

Medium

Approval. Agent proposes;
user reviews and decides

3

High

Monitoring. Agent runs;
user watches with
pause/stop capability

Error Communication and Recovery

Match the UI response to the type of error:

- Temporary / transient → auto-retry with progress: "Retrying... attempt 2 of 3"
 - User input error → clarify and educate: "Found 3 employees named John Smith — which one?"
 - Capability limit → explain and redirect: "I cannot process video — try VideoAnalysisBot"
 - Policy violation → explain the rule and show the approval path: "Deletion requires manager approval and backup confirmation"
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Worked Example

Downloading paper 1 of 20... 142ms

Extracting text... 2.3s

[... 198 more lines ...]

Token count: 15,847 API cost: \$0.47

Result: Attention mechanisms improve efficiency by 23%.

Improvements (hints)

1

Progressive
Disclosure

2

Transparency +
Control

3

Full Context
Awareness

Improvements

1

Progressive Disclosure

"Analyzed 20 papers. Key finding: 23% efficiency improvement (confidence 87%). [Show Reasoning] [View Papers]"

2

Transparency + Control

Real-time progress ("Processing paper 12 of 20, ~45 seconds remaining"), Pause and Stop buttons, confidence breakdown after completion, proactive note on conflicting results.

3

Full Context Awareness

Session header with conversation history, current analysis labeled in sequence, results structured from main finding → evidence → trade-offs → related questions, session save and export actions.

The HITL Challenge — Calibration

- The problem is not whether to include human control — it is where to set the intervention threshold.
 - Too much control → approval fatigue: Users blindly click "approve" without reviewing, defeating oversight.
 - Too little control → users feel powerless: Agents make consequential mistakes with no intervention point.
 - Effective HITL design matches intervention level to decision risk on a continuous spectrum.
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How to prevent
approval fatigue?

Preventing Approval Fatigue — Five Strategies

1. **Batch similar requests** — group 47 spam posts for bulk review with 3 examples, not 47 dialogs
 2. **Progressive detail** — show essential decision first, expand for full reasoning
 3. **Smart thresholds** — adapt auto-approval limits to observed user behavior
 4. **Keyboard shortcuts** — A to approve, R to reject, M to modify in high-volume queues
 5. **Time-based defaults** — "If no response in 2 hours, agent proceeds with recommendation" (for non-critical decisions — communicate clearly)
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Accessible Design — Beyond Compliance

WCAG 2.1 Level AA is the baseline, not the goal. Accessible design improves interfaces for everyone:

- Screen reader requirement → forces logical information hierarchy → better for all users
- Plain-language error messages for screen readers → clearer for everyone
- Text summaries of visual reasoning graphs → easier to copy into reports, scan on mobile

Four WCAG principles applied to agent UIs:

- Perceivability, Operability, Understandability, Robustness
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Challenges &
Successes?

Quiz review

- Great job overall
- Important points met
- More quiz takers are needed

Quiz Review

What distinguishes effective agent chat interfaces from simple message displays?

A:

- Feedback mechanisms and quick actions

Note:

- Quick actions: buttons for common follow-ups
 - What other feedback mechanisms that you have seen?
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Quiz Review

A loan approval agent rejected an application. You need to implement feature attribution to help the applicant understand the decision. The agent considered 8 factors, with these influence scores: - Credit score: 0.45 - Debt-to-income ratio: 0.30 - Employment history: 0.12 - Savings balance: 0.05 - Payment history: 0.04 - Age of credit: 0.02 - Recent inquiries: 0.01 - Property value: 0.01 What should the explanation show?

A:

- Top 4 factors with influence weights

Note:

- Listing all factors with no prioritization may obscures the key drivers
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Quiz Review

A job application agent rejected a candidate with this profile: - Years of experience: 3 (requirement: 5+) - Relevant certifications: None (preferred: AWS/Azure) - Education: Bachelor's degree (requirement: Bachelor's+) - Management experience: None (requirement: 2+ years).

How should you design counterfactual explanations to provide actionable guidance?

A:

- Show what would change the outcome: "With 2 more years of experience OR AWS certification OR management experience, you would qualify"

Note:

- Rejection explanation turns disappointment to a roadmap
 - Explanation should communicate the minimal changes that would flip the decision
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Quiz Review

You're designing a document analysis agent for three user personas: casual users who just want summaries, business analysts who need reasoning details, and compliance auditors who require full technical audit trails. How should you structure progressive disclosure to serve all three personas? (Select 2 correct answers)

A:

- Create three disclosure levels: essential view (summary + confidence), expanded view (key points + reasoning), technical view (full trace + citations)
- Hide technical information in collapsed sections to simplify the interface

Note:

- Document complexity should not dictate how analysis results are displayed.
 - You need to select 2 options in answering multi-select questions.
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Quiz Review

An agent approval system shows these metrics: - 98% approval rate - 2-second average review time - 200 approval requests per day per user - Recent audit found 22% of approved requests violated business rules - Users report feeling overwhelmed by approval volume - Current system: all requests above \$50 require approval, uniform confidence threshold of 70%. Diagnose the root cause of the approval fatigue and recommend a comprehensive fix addressing thresholds, volume, and user experience.

A:

- Miscalibrated thresholds sending too many low-value requests. Fix: (1) Implement risk-based confidence gates. (2) Add risk gates considering factors beyond amount.

Note:

- Best practices: understanding the root cause(s) of a problem first, then solve it comprehensively
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Quiz Review

You are designing a monitoring dashboard for a batch data processing agent that processes 200 customer records per hour, running 8-hour jobs overnight. The agent performs data validation, enrichment, and deduplication. What should the monitoring interface display?

A:

- Show aggregate real time status by default with aggregated metrics. Provide drill-down capability to examine individual failed records. Include prominent intervention controls: Pause, Stop, and Resume.

Note:

- Try to identify key scenario details that might greatly influence answer choices
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Next Chapters

- **Chapter 1.2: Core Patterns**

ReAct (Reasoning + Action), Plan-and-Execute (Hierarchical Task Decomposition), Reflection (Self-Critique and Iterative Refinement), and Tool-Use Architecture—examining their strengths, limitations, and production applicability.

- **Chapter 1.3: Multi-Agent Systems**

The fundamental challenge of coordinating multiple autonomous agents toward shared or competing objectives. It examines collaborative paradigms using specialized agents with shared goals, competitive systems applying game-theoretic principles, swarm intelligence emerging from simple local rules, communication mechanisms and orchestration patterns.